

JAVA

1. 1. Write a program for Multiplication table of 10 using any of the looping statement

```
package operators;

public class Table {

    public static void main(String[] args) {

        // TODO Auto-generated method stub

        int num =10 ;//table we want

        for (int i = 1; i<=10 ; i++) {

            System.out.println(num + "x" + i +"="+(num*i));

        }

    }

}
```

OUTPUT :

```
10x1=10
10x2=20
10x3=30
10x4=40
10x5=50
10x6=60
10x7=70
10x8=80
10x9=90
10x10=100
```

2.What is the difference between for loop and while loop ?

For Loop	While Loop
Used when the number of iterations is known in advance.	Used when the number of iterations is unknown in advance.
Initialization, condition, and update are written in one line.	Initialization is written before the loop, and update is written inside the loop.
More compact and easier to read for counting loops.	Slightly longer because initialization and update are separate.
Best for fixed-range loops (e.g., 1 to 10).	Best for condition-based loops (e.g., until the user enters a valid input).
Less chance of forgetting the update statement.	Higher chance of creating an infinite loop if the update statement is missed.
Commonly used for arrays and collections.	Commonly used for menus, input validation, and waiting for a condition.
All loop control statements are grouped together.	Loop control statements are spread across the program.
Variable can be declared inside the loop header, limiting its scope.	Variable is usually declared outside the loop and can be used after the loop.